NAME: ASHOK

REG.NO:192312558

Ex 8: Depth First Search (DFS)

CODE:

```
def dfs(graph, start, visited=None):
    if visited is None:
        visited = []
    visited.append(start)
    for neighbor in graph.get(start, []):
        if neighbor not in visited:
            dfs(graph, neighbor, visited)
    return visited


graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}
print("DFS Traversal:", dfs(graph, 'A'))
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-8.py ============
DFS Traversal: ['A', 'B', 'D', 'E', 'F', 'C']
```